

An 85-year-old lady is scheduled for hemiarthroplasty for a neck of femur fracture sustained after a fall. She has a background history of Alzheimer's dementia, hypertension, type 2 diabetes mellitus and ischemic heart disease.

i) Outline the innervation of the hip. (20%)

ii) Compare and contrast the use of a femoral nerve block with fascia iliaca block for analgesia in this patient. (30%)

iii) Justify strategies used to mitigate post-operative delirium in this patient. (50%)

i) Outline the innervation of the hip. (20%)

Sensory innervation of hip joint

Anterior capsule: Femoral nerve, obturator nerve.

Posterior capsule: Sciatic nerve, superior gluteal nerve, and nerve to quadratus femoris.

Motor innervation of hip joint

Flexors: Femoral nerve

Extensors: Sciatic nerve

Adductors: Obturator nerve

Abductors: Superior and inferior gluteal nerves

ii) Compare and contrast the use of a femoral nerve block with fascia iliaca block for analgesia in this patient. (30%)

Location / Targeted Nerves

- Femoral nerve block (FNB) directly targets the femoral nerve at the level of the inguinal ligament
- Fascia iliaca block (FIB) works injecting high LA volume below the fascia iliaca to achieve spread to femoral nerve, obturator nerve and lateral femoral cutaneous nerve (LFCN).

Localisation of nerve

- Nerve stimulator may be used for FNB to localise femoral nerve, but is not useful for FIB due to distance of injection nerve from femoral nerve.

Analgesia efficacy

- Good supporting evidence (RCT) for both FNB and FIB for post-op analgesia for hemi-arthroplasty.
- Both are effective at reducing post-op opioid requirement.

Drug Dose

- For the same LA concentration (ranging from 0.2% to 0.5%), FNB requires lower volume (15-20ml) compared to FIB (30-40ml) as the latter require high volume for LA spread within the fascial plane.

Onset of block

- For the same drug concentration, FNB has earlier onset of block action (within 15min) due to more direct LA deposition around the femoral nerve compared to FIB.

Duration of Action

- For the same drug concentration, FNB has shorter duration of action (6-12h) due to lower volume/ total dose.

Sensory Blockade

- FNB blocks femoral nerve which supplies part of the anterior capsule.
- FIB blocks femoral nerve and obturator nerves, which supply the anterior capsule. It also blocks the LFCN, which supplies the skin over the surgical incision on the lateral thigh.

Motor Blockade

- Both results in motor blockade of the quadriceps muscle (hip flexion), which may affect patient's post-op ambulation and rehabilitation.

Suitability for catheter infusion

- Both are appropriate for catheter siting and infusion.

Potential Risks

- FNB is associated with higher risk of direct needle tip injury to femoral vessels and nerve due to close proximity of injection from femoral nerve compared to FIB.
- As such, FNB is not appropriate to be done under GA due to close proximity to nerve while FIB is generally safe to perform when patient is under GA.
- FNB has lower risks of LA systemic toxicity (LAST) due to lower drug dose/ volume compared to FIB.
- Supra-inguinal approach to FIB is associated with higher risks of (1) bowel injury and (2) haematoma formation in the event of inadvertent vessel injury (not easily compressible) compared to FNB.

iii) Justify strategies used to mitigate post-operative delirium in this patient. (50%)

Strategies can be divided into pre, intra, and post-op.

Pre-op optimization

Identification of high risk patients

- Use of cognitive screening tools like MMSE/ Mini-COG/ MoCA
- High risk patients (like this patient) will benefit from additional preventive/ mitigating measures, and should be prioritised for early surgery to be done during office hours.

Pain control

- Optimise analgesia, including regional anaesthesia catheter as appropriate
- Drug dosing to be adjusted for renal/ liver impairment

Management and optimisation of co-morbidities

- Correct anaemia and coagulopathy
- Ensure electrolytes replete
- Optimise sugar control
- Optimise BP control in ward
- Review and minimise polypharmacy (and cognitive side effects)

Thromboembolic prophylaxis

- Calf pumps/ compressors
- Anti-coagulants for high risk patients, with appropriate cessation before surgery.

Minimise prolonged fasting time

- Plan surgery in elective trauma OT
- IV hydration when NBM

Pre-op education

- Opportunity to allay fears and address concerns
- Counsel patient and family on what to expect to reduce peri-op anxiety

Intra-operative strategies

Minimise sedatives and general anaesthetics

- Preference for CNB over GA if possible
- Avoid drugs which increase risk of delirium: ketamine, benzodiazepines, atropine
- Avoid deep plane of anaesthesia/ burst suppression for GA
- Titrate anaesthetics according to BIS target of 40-60

Opioid-sparing techniques

- Regional techniques for analgesia
- Multi-modal analgesia including paracetamol, COX-II inhibitors if no contraindications
- Short-acting opioids during painful stimulus (skin incision, bone manipulation, instrumentation) intra-op

Ensure adequate cerebral oxygenation and perfusion

- Close monitoring with regular NIBP or IA BP to maintain MAP within 20% of baseline (and > 65 mmHg)
- Maintain normovolaemia (run maintenance IV fluids and replace fluid losses)
- Keep SpO₂ > 95%
- Keep Hb > 8 g/dL

Keep patient warm

- Monitor intra-op temperature

- Passive (HME filter) and active (circulating air mattress) warming

Post-op strategies

Early orientation

- Recover in quiet environment
- Return visual (spectacles) and hearing aids
- Early orientation with visual and verbal cues (clocks, updates from nurses/ family members) once patient is awake.

Keep patient warm

- Blankets to keep patient covered in recovery.
- Active warming as appropriate

Early enteral feeds

- Adequate PONV prevention (dual anti-emetics, adequate hydration) if under GA
- Encourage orally as tolerated

Early mobilisation

- Motor-sparing regional techniques (pre/intra-op) where possible
- Involvement of PT/ OT/ family to start early mobilisation
- Adequate analgesia with breakthrough regime to facilitate participation
- Thromboembolic prophylaxis for patient who are unable to comply with early mobilisation

Minimise lines/ tubes/ drain

Stool softeners/ laxatives to facilitate bowel movement and avoid constipation

Review medications, minimise polypharmacy and avoid psychoactive medications

Based on current evidence, there is no single universal protocol for mitigating post-op delirium. Resources should be rationalised and individualised based on patient's profile.

Given that this is an elderly patient with significant risk factors (pre-existing dementia, immobility, pain, cardiovascular disease, major surgery) for developing post-op delirium, she should be prioritised for early surgery and abovementioned mitigating measures should be instituted where feasible.